S(21): DYADIC REDUCTION AND RELATIVE FORMAL IMMERSIONS

31 October 2026

Theorem. For elliptic curves over number fields of exact degree 21,
$S(21)=\{2,3,5,7,11,13,17,19,23,29,31,37,41,43,47\}$.
This is a computer-assisted proof. All required local certificates, selected-character families, independent replays, and the exhaustive prime and residue-support ledgers have passed. There is no unresolved case and no conjectural rational-torsion hypothesis.

1. OCCURRENCES AND WEIGHTED TRACES

The rational examples for 2,3,5,7 extend to a degree 21 field. For 11,13,17,19 the genera of $X_1(p)$ are 1,2,5,7. At a rational cusp c, Riemann--Roch gives $l(21c)-l(20c)=1$, since $20>2g-2$. A function with unique pole of exact order 21 gives a degree 21 map to P^1. Hilbert irreducibility supplies noncuspidal degree 21 points; the fine moduli interpretation supplies the elliptic curves and their torsion points. For 23 there is an explicit degree 7 example, extended by a cubic field extension. For 29,31,37,41,43,47 there are explicit degree 21 examples. verify_occurrences.m verifies irreducibility, nonsingularity, and the order of every explicit point. The data are in occurrences.json.

Conversely, suppose $p>47$ and (E,P) is defined over a degree 21 field. The corresponding point x on $X_1(p)$ has degree r dividing 21. We use
$D=(21/r)\,\mathrm{Tr}_Q(x)$.
It is a rational, effective, noncuspidal divisor of degree 21. Its geometric support is Galois transitive and its multiplicity is constant. The argument allows r=1,3,7,21 and nonreduced specializations.

The following observation is useful because 21 is odd. For $p>3$ the rational diamond group $G=(\mathbb{Z}/p\mathbb{Z})^*/\{+1,-1\}$ acts freely on $X_1(p)(R)$. Indeed, a fixed real moduli point has a real representative (E,P), since $\mathrm{Aut}(E,P)=1$. An isomorphism taking P to aP is unique and therefore real. But $\mathrm{Aut}_R(E)=\{+1,-1\}$, including at j=0 and 1728, so $a=+1$ or -1. Since an odd-degree point has a real conjugate and rational diamonds commute with Galois, $hD=D$ implies $\mathrm{ord}(h)$ divides r and hence divides 21. This also handles the CM cases in all stabilizer arguments below.

2. THE IMPROVED DYADIC BOUND

We use the dyadic winding theorem established in the preceding packets, in the integral form $2M_{\mathrm{lat}} \subset L \subset M_{\mathrm{lat}}$. Equality of these lattices is not asserted. Set m=42 and M=229. For every $1\le u\le 228$ the matrix
$(\varepsilon(na)-\varepsilon(nu/a))_{(1\le n\le 42,\,1\le a\le 228)}$
has rank 42 over $\mathbb{F}_2$. Arithmetic in the arguments is modulo 229 and ε is the characteristic function of residues $115,\dots,228$. prepare_bound.py produces all entries; verify_dyadic.py reconstructs and ranks them independently. The threshold is
$p>2\cdot 42\cdot 229=19236$.

The rational dyadic torsion reduction kernel has exponent 2. Doubling a degree 21 trace produces a degree 42 divisor with zero winding image. If its divisor polynomial before doubling is $1+2B(t)$, the polynomial after doubling is $1+4(B(t)+B(t)^2)$. The doubled point is therefore in the formal neighborhood modulo 4. The lattice inclusion supplies functions with linear terms $2x_i$ and integral terms of degree at least two. For a nonzero vector in $4\mathbb{Z}_2^{42}$, a coordinate of smallest valuation $k\ge 2$ has linear term of valuation $k+1<2k$; cancellation is impossible.

The formal zero fiber consists only of the cusp. Thus a trace whose specialization on X_0(p) is a multiple of one rational cusp is excluded, including ramified traces.

An ordinary elliptic curve over F_(2^f) has even group order. At f=21 its Hasse upper bound is 2100049; the largest possible even group order is 2100048. An odd prime divisor is therefore at most 1050024. Smaller ordinary residue degrees give smaller bounds. The complete supersingular group orders at the relevant degrees are

18: 261121,261633,262145,262657,263169;
19: 523265,524289,525313;
20: 1046529,1047553,1048577,1049601,1050625;
21: 2095105,2097153,2099201.

The last three are 5*29*14449, 3^2*43*5419, and 13*113*1429.
Every order at degrees 20 and 21 has all prime factors below 100000.
The full twist lists at 17,...,21 are verified natively and independently.
At 17 the orders are137*953,3*43691,5*26317, again with all prime factors below 100000. For f<=16 even the Hasse upper bound is below 100000.
Multiplicative identity components have orders 2^f-1 and 2^f+1.
Additive reduction cannot support these large odd primes.

For p>1050024 every dyadic place would therefore be multiplicative, with P on a nonidentity component. Its image on X_0(p) is the same rational cusp at every place. The improved dyadic theorem excludes this possibility. We obtain p<=1050024. Above 100000, testing every even multiple of p in each ordinary Hasse interval, all supersingular orders, and both toric orders for every f<=21 leaves 850 primes. No assumption that a residue degree divides 21 is made. The screen allows arbitrary partitions of 21 among the dyadic places.

3. GLOBAL DIAMOND EXCLUSIONS

We use Derickx--Stoll, Prime order torsion on elliptic curves over number fields, Part I: Asymptotics, arXiv:2505.14109v1, Theorem 3.3, Proposition 4.1, Table 2, Remark 4.4, and Corollary 7.5. Table 2 is used only in the proved range p<100000. The rational cuspidal 2-primary result is used only for p<=157. The gonality estimate is

gon_Q(X_1(p)) >= 325*(p^2-1)/65536.

Write str(p) for the order of the group generated by strange characters. If p^2>(2^19/325)*21+1 and (p-1)/(2str(p)) does not divide 21, Corollary 7.5 excludes p. Apply it to the exact degree r of the modular point, where r divides 21; the size condition holds because r<=21. All these r are odd, so its even-degree CM exceptions are absent. Together with Proposition 4.1 this leaves the following primes below 100000:

53,59,61,67,71,73,79,83,89,97,101,103,107,109,113,127,131,137,139,
163,167,181.

The first nineteen are treated locally below.

At 163 take h=diamond8, of order 27, and B=im(h-1). The frozen exact A_3 determinant norms on character orders 9,27,81 are odd, on full sign-zero rational homology of dimension 1980. Here

A_q=T_q-diamond(q)-q

is the Fricke-transported convention. Hence A_3 on B is an odd-degree isogeny. Since (h-1)J(Q) is finite and killed by A_3, this finite group has odd order. Proposition 4.1 with ell=2, n=6 applies: 6*21=126 is strictly below gon(X_1(163)). It gives hD=D, impossible since 27 does not divide 21.

At 167 use h=diamond2, of order 83. The inherited exact A_3 norm on the character-order 83 component is odd; its full rational homology has dimension 2132. The same argument gives the 2-primary hypothesis. Proposition 4.1 with n=5 gives invariance because 105<gon(X_1(167)); 83 does not divide 21.

At 181 take r=diamond2, h=r^6 of order 15, B=im(h-1). The only positive nontrivial character order in Table 2 is 6, so (h-1)J(Q) is finite. The full rational homology of B has dimension 2388, consisting of orders 5,9,10,15,18,30,45,90. For orders 5 and 10, A_11 has odd determinant norm; for all the other orders, A_3 does. These are recomputed with Sage modular symbols, independently of the native exported matrices. Put u=r^2, of order 45, and E=5*N_(u^5)-N_u, where N denotes the sum of all powers. On character orders 5 and 10, E acts as 45; on the other six orders it acts as zero. The integral endomorphism

 beta=E*A_11+(45-E)*A_3

kills rational torsion and restricts to an odd-degree isogeny on B. Thus (h-1)J(Q) has no 2-primary torsion. Proposition 4.1 with ell=2, n=6 applies since 126<gon(X_1(181)). It gives hD=D, impossible since 15 does not divide 21. No assertion about all rational torsion on J_1(163), J_1(167), or J_1(181) is needed.

4. COMPLETE SELECTED-CHARACTER FAMILIES

For each of the 850 large primes put m=(p-1)/2. Choose a prime ell with v=v_ell(m)>w=v_ell(21), and test every character order n dividing m and divisible by ell^(v-w). If every such order is nonstrange, the ell-primary part of str(p) has order at most ell^(v-w-1). Consequently m/str(p) is divisible by ell^(w+1), which does not divide 21. Corollary 7.5 then excludes p and every possible smaller field degree dividing 21. The ledger specifies all selected families: 5289 orders, of which 2474 are inherited and 2815 require fresh certificates. No characters above 100000 are supplied merely by extrapolating Table 2.

For a selected order n, work over a prime field F_q with q=1 mod(p-1) and a character of exact order n. The primary Manin presentation uses the S/I quotient, all three-term relations A, and the four-term T_2 operator R before quotienting by A. Proportional duplicate relations are normalized; each generator occurs in at most two distinct rows. The exact graph certificate gives both lower and upper bounds for rank(A), hence full relative dimension k+2. The auxiliary prime q is coprime to6p(p-1). After inverting these factors, the character summand of relative homology and its two-dimensional boundary commute with reduction. Thus k is also the characteristic-zero cuspidal dimension, and a full-rank winding witness modulo q lifts to characteristic zero. Every relation on the chosen functional, every prepared R row, and the winding vector are checked independently. Multiplying the relative winding class by

 (R-(2chi(2)+1))*(R-(chi(2)+2))

kills the two boundary factors. Thus the resulting scalar sequence factors through the actual cuspidal space of dimension k. This uses the cuspidal space, not the much larger raw generator count.

The initial raw engine is retained with its exact source and binary. Whenever its one-sum uint64 bound is inapplicable, blocked accumulation or the verified dual-coordinate engine is used. A safe raw bound is ns*(q-1)^2<=2^64-1. Blocked and dual engines use bounded partial sums or uint128 as documented in their sources. Every engine’s source, binary, actual log, and per-instance input are bound to the certificate.

The dual-coordinate engine uses ker(A) on the dual. Gauge relation rows and edge columns so forest edges have coefficients +1 at a child and -1 at its parent. Nonforest columns are free coordinates. Each unbalanced component has one additional pivot; leaf elimination and one root equation reconstruct all remaining coordinates uniquely. The number of free coordinates is k+2. Balanced roots have zero residual; dummy anchored roots impose no equation. The independent checker verifies every gauged incidence coefficient, all forest pivots and their order, every free operator row against raw R, the scaled initial functional and winding vector, and a full ambient one-step operator comparison. Since R preserves ker(A), applying its free rows and reconstructing equals the full R action. The scalar sequence is therefore unchanged. A real dimension-87500 example has also been replayed for all 175000 terms against the original raw sequence. Synthetic overflow and forest fixtures are additional software checks, not premises about a modular curve.

Each fresh certificate retains 2k scalar terms. Native prefix checks independently confirm the first 64 terms from the original Manin data. A recurrence is constructed either by native Berlekamp--Massey or by NTL half-gcd. Independently, Sage/FLINT verifies polynomials C,A with
 C(0)=1, deg C<=k, deg A<k, gcd(C,A)=1,
 C*S=A mod x^(2k), max(deg C,deg A+1)=k.
These equations prove minimal recurrence length k: a smaller fraction A′/C′ would make C′A-CA′ vanish modulo x^(2k), although its degree is less than 2k, so it would vanish identically; coprimality then contradicts the smaller degree. Full scalar complexity k proves that the cuspidal winding vector is cyclic. Thus the characteristic-zero winding projection on every relevant factor is nonzero. Galois conjugacy covers the entire selected character order. This is an exact certificate, not a probabilistic Wiedemann conclusion. verify_large.py checks every family, archive, hash, engine, prefix log, and complete Pade identity.

5. FINITE-FIELD SOURCES AND PRINCIPAL-DIVISOR OBSTRUCTIONS

At the nineteen local levels, census_sources.m enumerates every noncuspidal closed point of degree at most 21, by actual elliptic curves, all twists, and p-torsion lines. Proper j-fields are included. When full p-torsion is rational, all p+1 cyclic subgroups are included. The new full-torsion phenomenon at p=127, f=21 is explicitly covered. Ordinary counts are independently checked by class numbers, with the extra full torsion term p*H((t^2-4q)/p^2) when the full-torsion congruences hold. Supersingular counts are independently checked using quaternion Frobenius and the 24 automorphisms. At 127 all 21 cohomological traces are additionally recomputed using independent Sage modular symbols; they agree with the native counts and class-number census.

The degree 20 packet, including its degree 19 dependencies, is frozen by a 37906-file manifest. Every file has been streamed and hashed. All new canonical models are byte-identical to degree 20, except 61, which uses its frozen degree 19 model. Actual Tate-normal-form diamond orbit matches bind every old source to the new census. All ordinary degree 21 sources are fresh; at 61 ordinary degrees 20 and 21 are fresh. Lower sources and lower-total pairs are inherited only through those matches. No random point search is used as a completeness argument.

The canonical q-series and unit equations describe the actual smooth modular special fiber. Their full integral cotangent assertion is inherited: a minimal 2-denominator would yield a nonzero mod 2

differential vanishing past its geometric zero bound. No assertion that an optimal winding quotient has the full cotangent lattice is used. Every new ordinary record has actual 3-isogeny targets, weighted total degree four times the source degree, and checked curve, division, unit, and jet equations. Repetitions and collisions use the required jets.

For the elliptic correspondence use A_3=T_3-1-3*delta3. The finite-image annihilators alpha are
 1 at 59,71;
 sigma^6-1 at 61;
 (sigma-1)*(sigma^6+1) at 97;
 (sigma-1)*(sigma^5+1) at 101;
 sigma-1 otherwise.
Table 2 proves the nontrivial character condition; the actual full Jacobians at 59 and 71 have rank zero. For an odd good prime q, T_q-1-q*deltaq kills rational torsion: rational torsion injects under reduction at q, and Eichler--Shimura kills its reduction. Thus alpha*A_3 kills the relevant rational divisor classes. Rational cusps contribute zero. On the q-form model its Fricke transport is T_3-D3-3 up to an invertible diamond factor.

For a resulting divisor U-V over F_2, use L=K^2 at 53,61,67 and L=K elsewhere. Choose nonzero f in H^0(L(-U)). If U-V=div(v), then f/v is a nonzero section of L(-V); for every h in H^0(L(-U)), h*(f/v) belongs to f*H^0(L). Exact multiplication tensors and the jet conditions give necessary linear equations on f/v. A matrix of full column rank forces it to be zero, contradicting principality. The actual f,h witnesses and matrices are retained. Randomness finds witnesses; verification consists entirely of exact identities and ranks. For every product tensor the retained precision P satisfies P-2>4g-4 for products in 2K, or P-4>8g-8 for products in 4K. Thus the q-series determine all multiplication identities globally by the geometric zero-divisor bound. In particular, at 67 precision 1300 gives: products lie in 4K of
degree 8g-8=1232 and have vanishing order at least1296 if all retained q-coefficients vanish. A weight8 Sturm bound is not claimed there.

There is one explicit exception to the ordinary principal-divisor tests: at 61, source 2921 has degree 20 and closed diamond orbit 6. It is a full H5 pullback and is annihilated by sigma^6-1. The strong relative maps have base rank 16 in degree 20, with rank 17 after adding any rational cusp. All six translate configurations pass and are independently rebuilt. The order 5 action is free, so the zero family is a pullback of a degree 4 divisor on the genus 26 quotient, plus the fixed cusp C1. Thus it cannot contain the noncuspidal divisor D. This is the only exception allowed by the final ordinary-source coverage guard.

Fresh ordinary pairs cover every relative diamond position of total 21; at 61 totals 20 and 21 are fresh. At 59,61,67,71,73 all unordered 7+7+7 triples, including repetitions and shorter diamond orbits, are covered. Independent Burnside counts prove completeness and distinctness of the triple representatives. In total the new pair/triple batches contain 35631 full-rank matrices. New delta2 mixed tests cover O12+NC9 and O15+SS6 at 73, O10+NC11 at 89, O12+SS9 at 109, O14+SS7 at 113, and O14+NC7 at 127. Lower mixed tests are inherited, since adding any delta2-invariant divisor leaves that obstruction unchanged. This also covers O7+2SS6 at 73. Its O7+O7+SS6 case uses the inherited19-position mixed pair certificate.

6. INTEGRAL ABSOLUTE AND RELATIVE FORMAL IMMERSIONS

The q-expansion computations use the dual-level model with a specified embedding of mu_p, whose Tate q-expansion cusp is rational. Sending (E,P) to E/<P> with the canonically identified dual kernel gives a Q-isomorphism from the point-level model to this model. This is the Fricke transport; it inverts diamonds and sends the point-level A_q factor to T_q-D_q-q up to an invertible diamond. Thus the usual q-expansion pairing with T_j is at a rational cusp in our conventions. The quotients whose cotangent lattices are used have odd orders 3,5,or7. Their tame quotients
of the good modular model at 2 are smooth, and averaging gives the saturated invariant differential lattice. The H8 pullback family at 113 is formed on the formal neighborhoods of a free ordinary orbit. Its local quotient is etale even at 2, so the family is smooth of dimension 1. No global even-order quotient cotangent assertion is required.

Pure cusp tests use the integral Hecke order, not an equality of Betti and Hodge lattices. For a prime-level weight2 space, q-expansion duality identifies the actual Z_2 cotangent lattice with Hom_Z2(T,Z_2). If t lies in 2T, its action on any integral T-stable lattice lies in 2End(L). Consequently operator independence modulo 2 on such a lattice implies the required independence in T/2T. This is the only direction used. Injective compression is verified by reconstructing every raw operator row. Exact quotient polynomials have finite image because their J0 images lie in the winding space and all positive nontrivial character orders are absent from the active quotients. The finite-image polynomial records remain bound to the frozen inputs.
The cusp argument uses only the full curve or quotients of odd order. It is independent of cotangent descent for an even-order quotient.

The coarse cusp checks at 71 and 127 fail in degree 21 and are not proof premises. They are replaced by the full canonical maps: A_q at 71, where J_1(71)(Q) is finite, and (D-1)A_q at 127. The available q in {2,3,5,7} are used. Their complete profiles pass (12360 and 7875). At 53 and 101 the full-curve repairs are likewise used. Full canonical maps also replace the even-order quotient tests at 61,89,97. At 89 use (D-1)Aq. At 97 use the same sharper Hecke factors as the SS16 map, with D-1 in place of D^16-1. At 61 the unique strange order 6 component has diamond2 eigenvalue z, z^2-z+1=0, and Hecke eigenvalues a2=-1-z, a3=-2, a5=-3+3z, a7=-2-2z.
The maps (D-1)*(Tn-an(D))*Aq have finite rational image: an exact winding-space containment certificate on the order 6 component, independently ranked over Q(z), proves that Tn-an(D) kills its sole non-winding factor; D-1 kills J0. Odd outer q kills rational torsion. All 8295 full cusp profiles pass. At 67 and 73 the complete cap6 search and every partition with a multiplicity at least7 are checked, including7+7+7. At the other levels, complete relation
spaces on dominating profiles prove prefix independence. At 103 and 131 only the known full simple cusp fibers remain; the intrinsic quotient relative ranks are20 in all 4845 and 125970 configurations respectively.

The relative formal lemma is integral. Let S be smooth of dimension d over Z_2 and L a smooth closed formal family of dimension r. Suppose an actual integral map Phi to a smooth group scheme vanishes on L and has normal cotangent rank d-r. Functions pulled back from the target lie in the ideal of L and have an invertible matrix of linear normal coefficients. They generate that ideal formally. Thus the zero locus is exactly L. This uses neither integration denominators nor an assumed

saturation for an optimal quotient. Closed-point coefficient matrices
and Serre trace-pairing matrices differ by an invertible basis change,
also for even residue degrees. Jets on symmetric powers require no
factorials. Canonical function jets and differential jets differ by a
common unit, giving an invertible triangular change.

All maps here are actual endomorphisms of the full Jacobian, or their
odd-degree quotient analogues. The usual odd q torsion annihilators
make rational images zero. For q=2, the proved fact that rational
2-primary torsion is cuspidal, for the local primes<=139, suffices:
A_2 kills that cuspidal subgroup; odd-primary rational torsion injects
under dyadic reduction. On the order 7 quotient at 127 the pullback has
odd kernel and is injective on 2-primary torsion, so the same assertion
descends. We never infer this assertion above 157.

A formal zero family consists of pullback fibers, tame fixed-point
lifts, and a fixed residual cusp divisor R. A nonzero R contradicts
the noncuspidal support of D. Complete binary relation spaces certify
that there are no other normal relations. To cover all multiplicities, move a maximal one
to cusp0. At most
floor(c/(cap+1)) entries exceed cap. Add the unused degree to the first
maximal entry so that these high entries sum to c. Enumerating all
resulting partitions and assignments therefore dominates every actual
multiplicity vector. Independent integer-partition and placement code
checks equality with the full sets of recorded profiles.
The cost of a jet relation
is the sum of prefix lengths, not its Hamming weight. The independent
Sage replay checks every full kernel and every unexpected coset.

The standard cases, written (base degree,base rank,cusp degree), are
 53 SS13: (13,12,8); full cusps53: (0,0,21);
 61 SS15: (15,12,6); new SS20: (20,16,1);
 61 ordinary20, source 2921: (20,16,1), all six translates;
 73 SS6: (6,6,15); NC9: (9,8,12);
 97 SS16: (16,16,5), with the inherited sharper finite maps;
 101 full cusps: (0,0,21), with the inherited sharper finite maps;
 109 SS9: (9,8,12);
 113 ordinary8: (8,7,13).
Available q from {2,3,5,7} are used for standard maps. The sharper maps
retain their odd outer torsion factors. Every entry is reconstructed
independently from the actual canonical q-series, Hecke action,
elliptic point, and diamond matrices.

At 73 double SS6+C9 has ten unordered translate configurations. The
new cotangents have base rank 12. If C9 is a full H9 fiber, no residual
cusp remains, but the zero family still contains the unique tame
H3-fixed SS lifts. Then D contains an h3-fixed noncuspidal point.
Galois transitivity would give an h3-fixed real point, impossible by
Section 1. Otherwise a fixed cusp remains. The SS6+NC9+C6 case has all
sixteen relative positions, base rank 14, and a fixed nonzero cusp
remainder. The optional triple-SS jet checks are not needed: total
nonordinary degree 18+C3 and degree 21+C0 are already excluded by the
H9 stabilizer, as in Section 8 below. The new SS73 jets and all combined
entries are independently rebuilt. Actual canonical point orbits are
also checked independently: in particular H9 is free on SS9 at 109,
H5 is free on both SS15 andSS20 at 61, and the claimed tame order 3
stabilizers at 73,97,127 fix their geometric points. Thus the stated
formal pullback and fixed-point families have their asserted dimensions.

7. THE ORDER7 QUOTIENT AT127 AND THE NEW SUPERSINGULAR TESTS

The direct supersingular tests use the same exact principal-divisor criterion. At 113 the SS7 source is nonprincipal under alpha*A3; at 109 the SS12 source is nonprincipal. At 127 there are six SS14 source orbits, with source indices20,...,25 and closed diamond orbit sizes 9,9,9,3,9,9 (the order is determined by the source file). T3 treats 20,21; T5 treats22,25; T7 treats24. The T5 calculation uses all six quadratic kernel factors; the T7 calculation uses all eight cubic kernel factors. The exceptional orbit 23 is handled on a quotient below. Every needed supersingular deformation varies a1 by u. The division polynomial derivative, curve equation, all modular unit equations, and a nonzero first projective jet are checked, so u is a uniformizer. Sage independently verifies these identities and reconstructs the isogeny target orbits, including extension-field kernels.

Let Y=X_1(127)/H7. The order 7 action is free: geometric elliptic stabilizers have order dividing24 at 2 and dividing6 in characteristic zero, and the cusp action is free. The quotient is smooth and has genus 88. Its actual cotangent lattice is the invariant sublattice: 7 is a2-adic unit, so averaging gives saturation. The exported invariant basis V has rank 88 and satisfies V*D^9=V; an independent kernel dimension check proves it is the entire invariant space.

For one, two, or three NC7 components, the h7 stabilizer argument has 2*(21+c)<=70<gon(X_1(127)). It forces D=pi7^*D′ of degree 3 onY. The special fiber D′ consists of three cusps among eighteen F_2-rational cusps: nine rational cusp images and nine nonrational cusp images. Frobenius2 belongs to H7. Use onY the maps induced by

$$(D-1)*(Tq-Dq-q)*(Tq-1-qDq),\ q=2,3.$$

The two Eisenstein factors annihilate both cusp types; their product is needed because the nonrational cusp lifts are not globally Q-rational. The first also handles the finite rational image. All 1140 unordered length-three cusp multisets, including repetitions, have rank 3. The NC expansions come from the Tate curve at a primitive127th root. Their curve and unit equations, unique normalized canonical jets, quotient projections, and every matrix entry are independently rebuilt. Absolute formal immersion then forces a cusp in D′, impossible.

For SS14 orbit 23 plus C7, h7 invariance follows from56<gon(X_1(127)). The quotient specialization is SS2+C1. Orbit23 is pointwise fixed by D^21 upstairs; the induced h3=D^3 onY is tame of order 3 and fixes its SS2 image. Use V*(D^3-1)*(Tq-Dq-q), q=2,3. All three SS translates and nine rational cusps give rank 3: twenty-seven disks. The zero formal family is the unique h3-fixed SS2 lift plus the fixed cusp C1. Thus D′ contains a cusp. Sage independently reconstructs all entries, allowing the harmless common unit used to normalize quotient forms.

For SS14+NC7, no cusp remains. The actual elliptic correspondence on NC satisfies T3=3+delta3, so A3 contributes2*(1-delta3). Add this term to the SS principal-divisor test. There are5*9+3=48 relative positions; all give nonprincipal divisors and full-rank 610 matrices.

For the exceptional ordinary9+NC7+C5 disk upstairs, the base degree is 16, its rank is14, and the known zero family has two pullback parameters. All 63 rational cusps and five jets are retained. Modulo the base, the complete315-column kernel has dimension 75. At each jet level it is exactly the 7-by-9 row/column code, generated by the H7 and H9 fibers. A row/column codeword has weight9a+7b-2ab; its minimum

nonzero weight is7. Different jet levels have disjoint coordinates. Consequently every five columns are independent. This proves normal rank 14+5=19 for all C(67,5)=9657648 cusp configurations. No cusp normalization is made after fixing the two nonrational supports. The fixed C5 cannot disappear. Hamming weight is used only for this code bound; general jet profiles use prefix cost.

8. STABILIZERS AND EXHAUSTIVE COVERAGE

Suppose the nonrational special divisor Z is h-invariant and C is a rational cusp lift of the complement c. The torsion class

$$y=[h(D-C)-(D-C)]$$

is well defined even when D-C has nonzero degree. It is torsion: if h has orderm, $m*y=(h-1)[m(D-C)-N_h(D-C)]$, with the bracket a rational degree-zero class in the finite image. Its dyadic reduction is zero, so2y=0. If gon>2*(21+c), this forces hD=D; cuspidal and noncuspidal parts cannot cancel. At 73 the independently verified actual $J_1(73)[2]$ upper bound is fixed by delta2. Since delta2 has odd order 9, y=0 there, so only gon>21+c is required. At 89 the frozen odd-degree anti-H11 isogeny likewise gives y=0: since11y lies in the anti-Jacobian and 2y=0, oddness of 11 puts y itself in its rational2-torsion subgroup. Neither uses an optimal-lattice assumption.

This handles nonordinary total 18 or21 at 73; NC11 at 89; nonordinary total 18 or21 at 109; total 14 or21 at 113; exceptional ordinary9 and its double at 127;
and SS17 or exceptional ordinary17 at 137. The even stabilizers at 109, 113,137 are immediately impossible by Section 1. At 73,127,137 the relevant odd orders 9,9,17 do not divide21. The SS7+ordinary8 and ordinary8+ordinary8 exceptional disks at 113 are handled by even H2 andH8 invariance, respectively. At 127 the allowed order 7 case is pushed to the smaller symmetric power onY, as detailed above.

The census gives477 possible multisets of ordinary degrees, supersingular degrees, and nonrational cusp degrees, with rational cusp complement. verify_coverage.py independently regenerates them and assigns a proved certificate or stabilizer to each. It also regenerates all 82136 primes through 1050024, checks every published reduction in its valid range, verifies every selected-character divisor family, and partitions the remaining primes by the complete dyadic screen. No unlisted residue type, source, relative diamond position, character order, or prime is allowed to pass the final guards.

The final_coverage.json ledger has passed, with status PROVED_WITH_COMPUTER_ASSISTANCE and no unresolved primes. All required independent source, rank, profile, character, and prime coverage checks have passed. This proves the theorem.

Primary reference:
M. Derickx and M. Stoll, Prime order torsion on elliptic curves over number fields. Part I: Asymptotics, arXiv:2505.14109v1 (2025), https://arxiv.org/abs/2505.14109v1 .
All reused computational premises are pinned by the degree 20 manifest; all degree 21 premises are listed in S21_certificate_index.txt.